

De Rham Cohomology

1.1 The De Rham Cohomology Groups

Definition 1.1.1

Let M be a smooth manifold with or without boundary, and let p be a nonnegative integer. Because $d : \Omega^p(M) \rightarrow \Omega^{p+1}(M)$ is linear, its kernel and image are linear subspaces. We define

$$\begin{aligned}\mathcal{Z}^p(M) &= \text{Ker}(d : \Omega^p(M) \rightarrow \Omega^{p+1}(M)) = \{\text{closed } p\text{-forms on } M\}, \\ \mathcal{B}^p(M) &= \text{Im}(d : \Omega^{p-1}(M) \rightarrow \Omega^p(M)) = \{\text{exact } p\text{-forms on } M\}.\end{aligned}$$

By convention, we consider $\Omega^p(M)$ to be the zero vector space when $p < 0$ or $p > n = \dim M$, so that, for example, $\mathcal{B}^0(M) = 0$ and $\mathcal{Z}^n(M) = \Omega^n(M)$.

Definition 1.1.2

The fact that every exact form is closed implies that $\mathcal{B}^p(M) \subseteq \mathcal{Z}^p(M)$. Thus, it makes sense to define the ***de Rham cohomology group in degree p*** (or the ***p th de Rham group***) of M to be the quotient vector space

$$H_{\text{dR}}^p(M) = \frac{\mathcal{Z}^p(M)}{\mathcal{B}^p(M)}.$$